Mathematics - Graduate Level Assessment 20 Total Marks: 40 Answer all questions.

Multiple Choice (10 marks)

1. If m and n are odd integers, how many terms in the expansion of $(m+n)^6$ are odd?
   (a) 6
   (b) 3
   (c) All the terms are odd
   (d) 4
   (e) 1
2. If f is a linear function with nonzero slope, and c < d, which of the following must be FALSE?
   (a) f (d) > f (c)
   (b) f (c) < f (d)
   (c) f (d) ≠ f (c)
   (d) f (c) = f (d)
   (e) f (d) < f (c)
3. Statement 1 | Every solvable group is of prime-power order. Statement 2 | Every group of prime-power order is solvable.
   (a) Statement 1 is always true, Statement 2 is false only for certain groups
   (b) True, True
   (c) Statement 1 is always false, Statement 2 is true only for certain groups
   (d) True, False
   (e) Statement 1 is true only for certain groups, Statement 2 is always true
4. Solve the equation -6 x = -36. Check your answer.
   (a) 42
   (b) 30
   (c) -42
   (d) -36
   (e) 1 over 216
5. What is the order of group Z_{18}?
   (a) 30
   (b) 15

(c) 12
(d) 6
(e) 24

True / False (10 marks)

Answer True or False

6. True or False: Calculating 60% of 30 yields the result of 18, which is a common mistake in percentage calculations.

7. True or False: The element c = 2 in Z_3 ensures that Z_3[x]/(x^3 + x^2 + c) is a field.

8. True or False: The polynomial 8 x^3 + 6 x^2 - 9 x + 24 is irreducible over Z but does not satisfy the Eisenstein criterion.

9. True or False: The covariance function C_N(2, 4) for a Poisson process with rate = 4 is 12.

10. True or False: The solution to the equation v - 26 = 68 is v = 94.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Find the roots of
$$6x^4 - 35x^3 + 62x^2 - 35x + 6 = 0.$$
Enter the roots, separated by commas.

12. Discuss how the amount of $20.08 can be represented as a fraction, and analyze the implications of this representation in mathematical contexts.

End of Paper